

Abstract – Hostel Management System

The project is designed to simplify the process of managing hostel activities by allowing administrators to manage student records, room allocations, fee details, and hostel operations through a web application without manual paperwork. The system aims to automate the traditional hostel management process, reducing time consumption and improving accuracy in room allocation and student management. It helps administrators maintain student information, room availability, fee payments, complaints, and attendance records in an organized manner. The project also enhances student convenience by providing easy access to hostel information, room details, fee status, and complaint services through the internet.

Modules:

1. Administrator

- Manages the overall hostel system
- Adds, updates, and removes room details
- Monitors student records and hostel activities
- Maintains room availability

2. Students / Users

- Register and log into the system
- View available rooms and hostel details
- Apply for room allocation
- Fee payment ,view fee status and submit complaints

3. Hostel Staff / Wardens

- Maintain student attendance records
- Manage room allocations and hostel facilities
- Handle student complaints and requests
- Monitor hostel activities and fee status.

Main Functionalities

- User dashboard for registration and secure login
- Room listing with availability details
- Online room allocation and management
- Fee payment and transaction management
- Complaint and feedback management
- Attendance management system
- Admin dashboard for managing rooms and students
- Staff dashboard for monitoring hostel activities
- Responsive frontend interface
- REST API integration between frontend and backend
- Secure authentication

The endpoints are commonly used in hostel management applications for handling authentication, room management, student records, complaint handling, and payment operations. These endpoints form the core communication layer of the Hostel Management System and contribute significantly to providing a responsive, efficient, and user-friendly hostel administration platform.

API Endpoints Included

- POST /api/register/ – Student registration
- POST /api/login/ – User authentication
- GET /api/rooms/ – Retrieve available rooms
- GET /api/rooms/{id}/ – View room details
- POST /api/allocations/ – Apply for room allocation
- GET /api/allocations/ – Retrieve allocation history
- PUT /api/allocations/{id}/ – Update room allocation
- DELETE /api/allocations/{id}/ – Cancel allocation
- POST /api/complaints/ – Submit complaints
- GET /api/students/ – Retrieve student details
- POST /api/payments/ – Handle fee payment processing
- GET /api/admin/dashboard/ – Admin management dashboard

Technologies Used

- Django
- React
- PostgreSQL
- HTML
- CSS
- JavaScript
- Bootstrap

The primary objective of this project is to develop a scalable and modern hostel management solution that fulfills the growing need for digital hostel administration services. The project demonstrates the practical implementation of full-stack web development concepts, authentication systems, REST API integration, database management, and software engineering principles in a real-world application environment within Software Engineering.